SRS: Data Source

1. Overview

The data source is the first part of the data pipeline. Data is read from the OutSystems QS-Connect database through two views, **CPAR** and **AP** (audit plan). The DBA team creates the views from SQL scripts provided by the engineering team, so the pipeline never reads the underlying tables directly. The pipeline connects with the DEV account provided by the DBA team. Creating the database, managing the account, and everything downstream of the source are outside the scope of this document.

2. Source Schema

All columns are nullable unless noted otherwise in the view definition. nvarchar(max) is shown where the source lists length -1.

CPAR view (56 columns)

Columns	Type
CPARId	bigint
AuditPlanId, Year, SequenceNumber	int
CPARNo	nvarchar(100)
AuditType, ForType, TypeOfNC, MinorNC, CauseCode, RootCause, SecondLevelRootCause, OFI, MilitaryFinding, Status	nvarchar(50)
RiskCategory, Company, Title	nvarchar(500)
QSSINo	nvarchar(1000)
LOB, Department, Programme	nvarchar(250)
Division	nvarchar(200)
Auditee, Auditor, Assignee, HOD, QM, DGM, GM, PM, QualitySupervisor	nvarchar(100)
Observation, RequirementDetails, Reason, RootCauseDetails, ImmediateCorrectionTaken, CorrectivePreventiveActionIm, VerifiedEvidence, LateVerifyReason, HOD_OverdueExplanation	nvarchar(max)
CreatedOn, LastUpdatedOn, DueDate, VerificationDuedate, ConcuredDate, CompletedDate, VerificationDate, ApprovedDate, EndorsedDate, NotifiedDate	datetime
IsDelete, IsVoid, IsMigratedData, IsOverdueConcurrence, IsOverdueVerification	bit

AP view (31 columns)

Columns	Type
AuditPlanId	int
AuditReportNo, Military, Customer, Authority, Certification, ServiceCenter, Tier, RiskCategory	nvarchar(500)
AuditType, ForType, Stage	nvarchar(50)
LOB, Department, Programme	nvarchar(250)

Columns	Type
Division	nvarchar(200)
LeadAuditor, HOD, QM, DGM, GM	nvarchar(100)
DateCreated, DatePlannedAudit, DateAgreedStart, DateAgreedEnd, DateActualStart, DateActualEnd, DateAuditComplete	datetime
IsPlanned, IsLocal, IsDisabled	bit

3. Functional Requirements

Requirement	Rationale
The pipeline must read data only from the CPAR and AP views in the OutSystems QS-Connect database, not from the underlying tables.	Views give a stable, controlled interface and keep the pipeline independent of internal table changes.
The engineering team must provide the SQL script for each view, and the DBA team must create the views from those scripts. Scripts must be versioned and linked to a request.	Keeps view definitions controlled, reviewable, and traceable.
The CPAR view must link to the AP view through `AuditPlanId`.	Allows each CPAR record to be associated with its audit plan downstream.
The pipeline must connect using the DEV account provided by the DBA team, with read-only access limited to the two views.	Applies least privilege and keeps access controlled by the DBA team.
Credentials must be stored in a secrets manager and must never appear in code, configuration files, or logs.	Prevents credential leakage.
The pipeline must fail clearly and raise an alert when the connection fails, a view is missing, or the schema does not match.	Source problems are noticed early instead of silently producing bad data.

4. Acceptance Criteria

- The pipeline successfully connects to the QS-Connect database using the DEV account and reads only from the CPAR and AP views.
- Both views exist and were created from the versioned SQL scripts provided by the engineering team.
- The columns and data types returned by each view match the schema in section 2.
- The DEV account has read-only access and cannot read other objects or modify data.
- Credentials are retrieved from the secrets manager and do not appear in code, configuration, or logs.
- A connection failure, missing view, or schema mismatch causes the run to fail with a clear error and an alert.